



Building AI Agents on the Akamai Cloud

What an agent is, how to build one, and how to run it on Akamai GPU and CPU nodes.

Du'An Lightfoot

Developer Advocate, Akamai · @labeveryday



Agenda

01 What is an agent

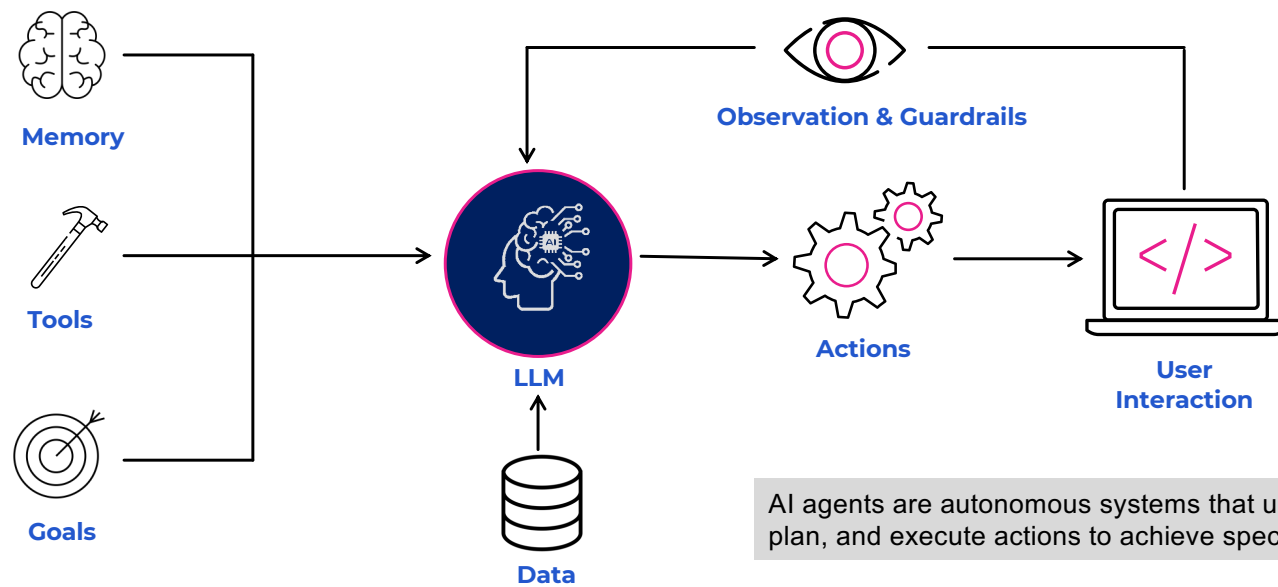
02 Components an agent

03 Building agents on Akamai Cloud

04 Demo

05 Resources

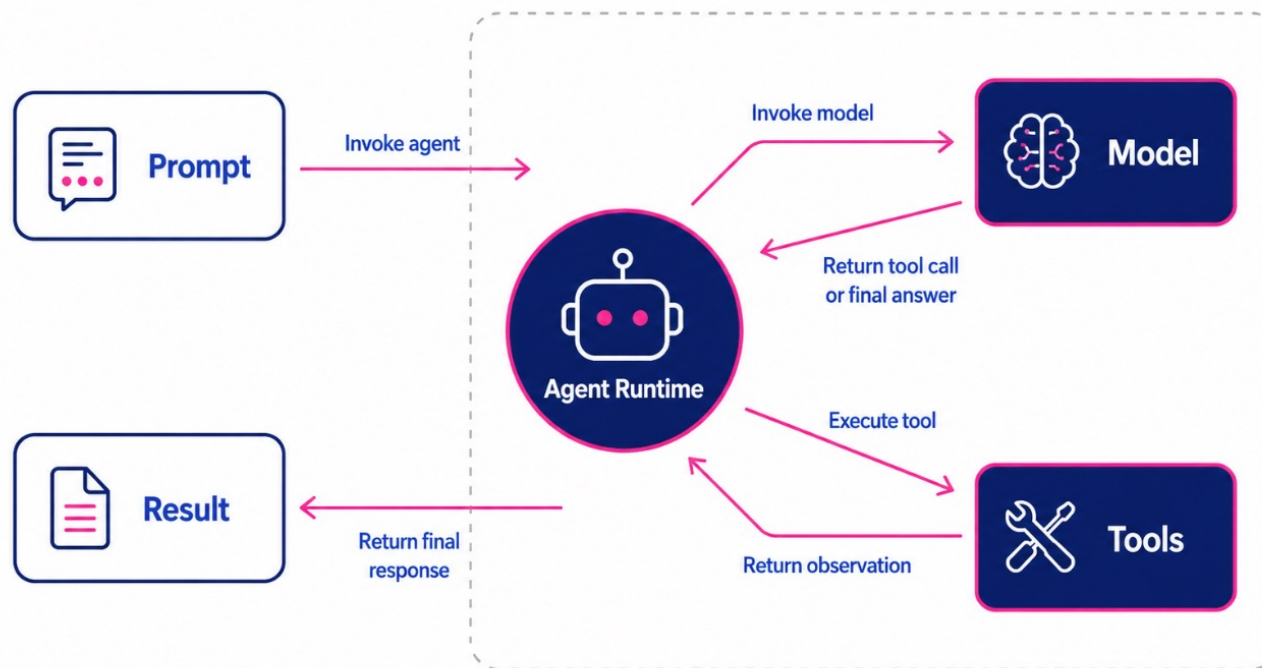
What is an agent?



AI agents are autonomous systems that understand, plan, and execute actions to achieve specific goals.

The agent loop

How agents operate under the hood



Agentic landscape

The tools are ahead of the guardrails

Utility Agents

CLAUDE
CODE

CODEX

Autonomous Agents

OpenClaw

NVIDIA
NemoClaw

Custom Agents

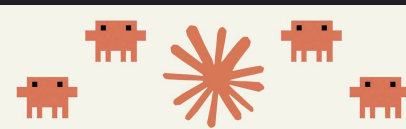
crewai

LangGraph

STRANDS
AGENTS SDK

Agent
Development
Kit

Managed Agents

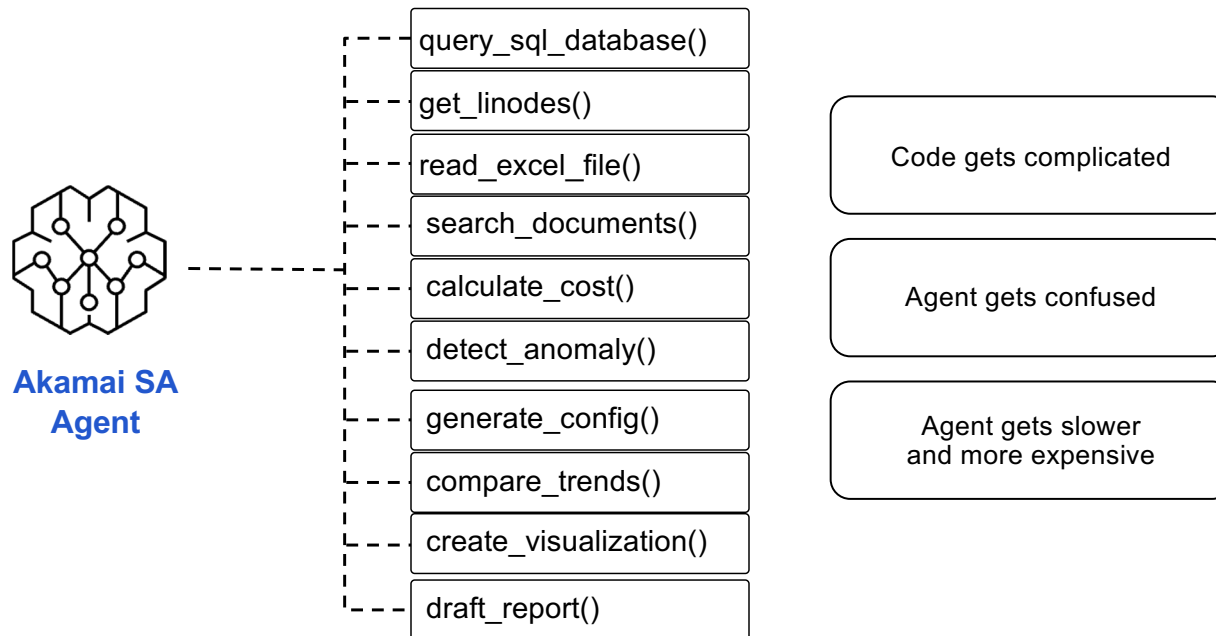


Claude Managed Agents

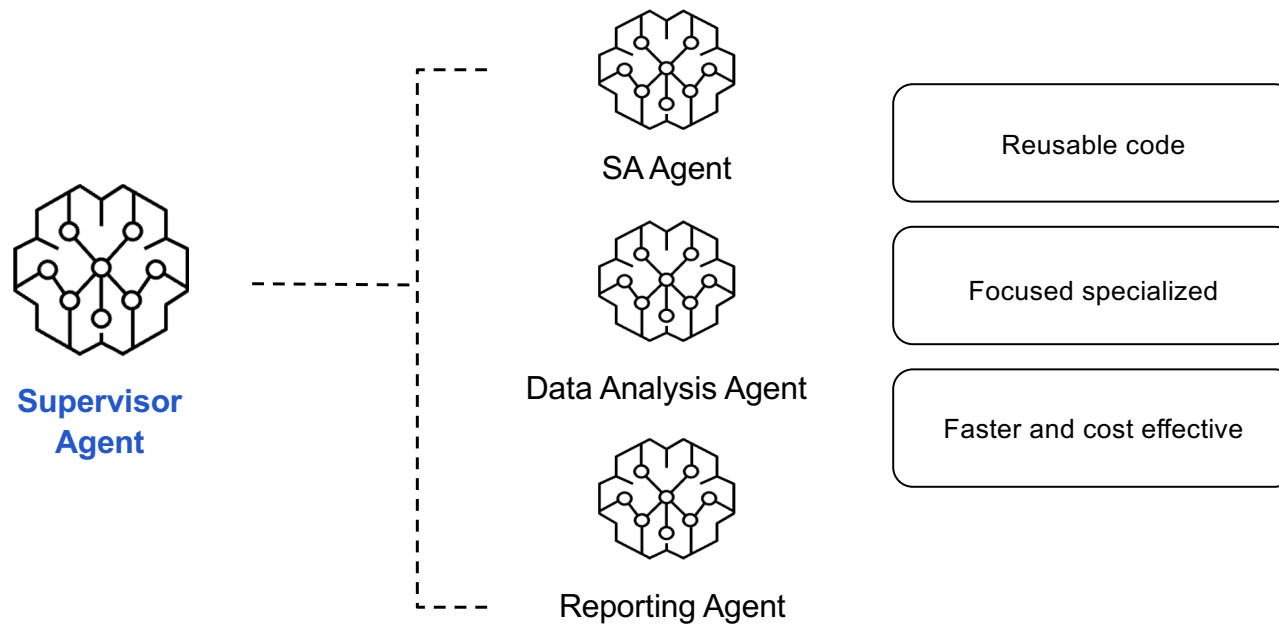
Agent Architectures

Understanding the basics of how agents are designed

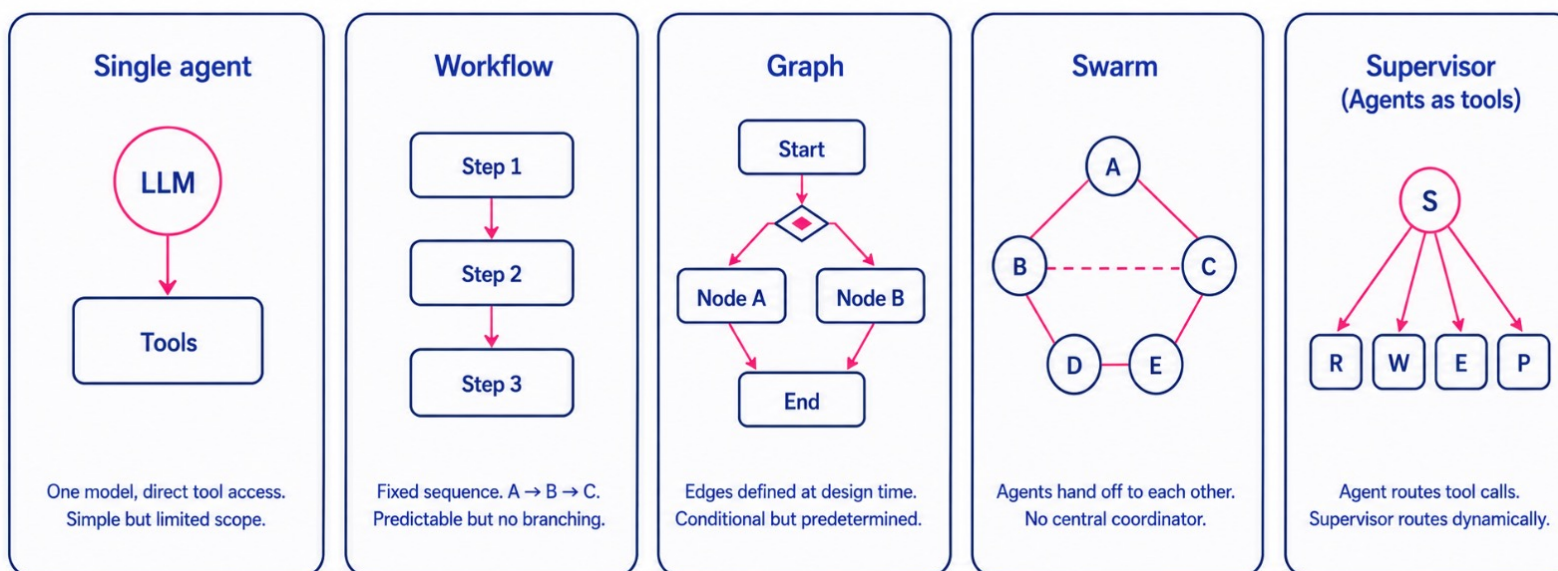
Single Agent



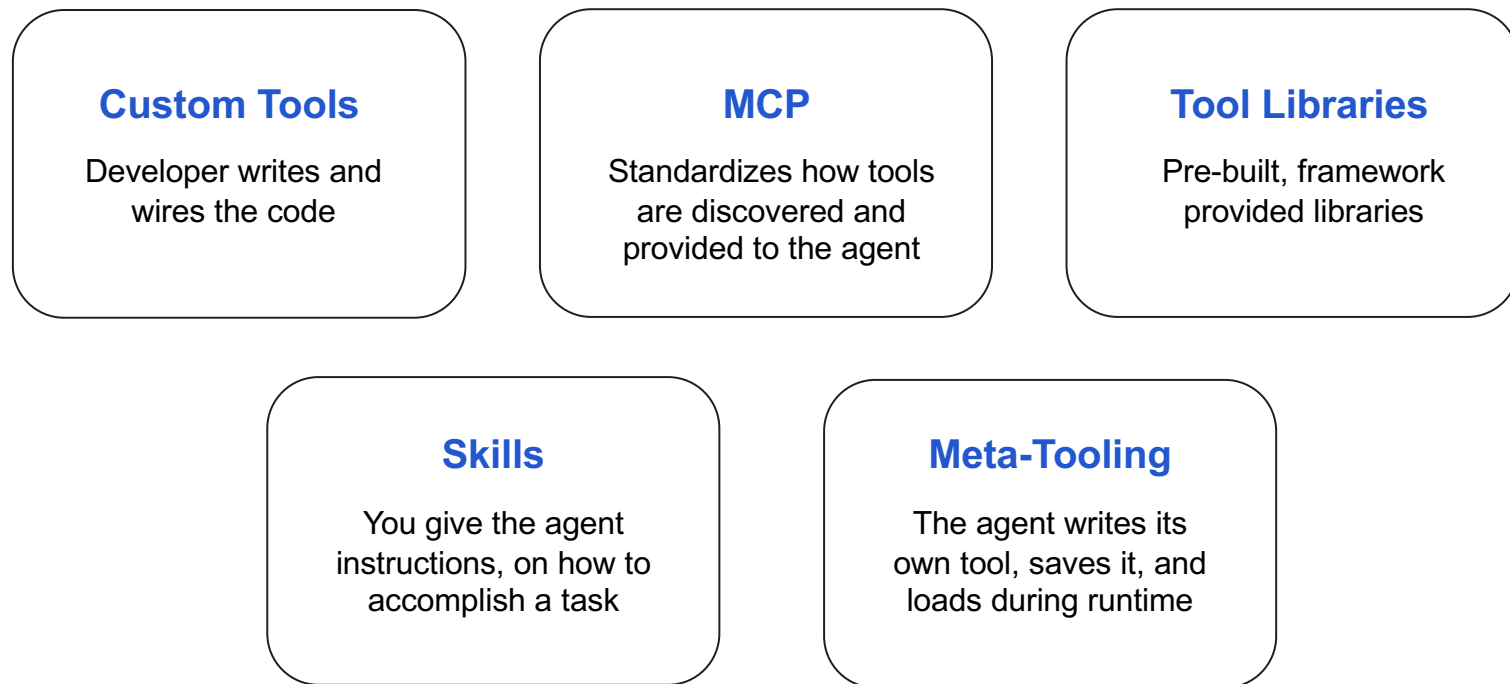
Multi-Agent Collaboration



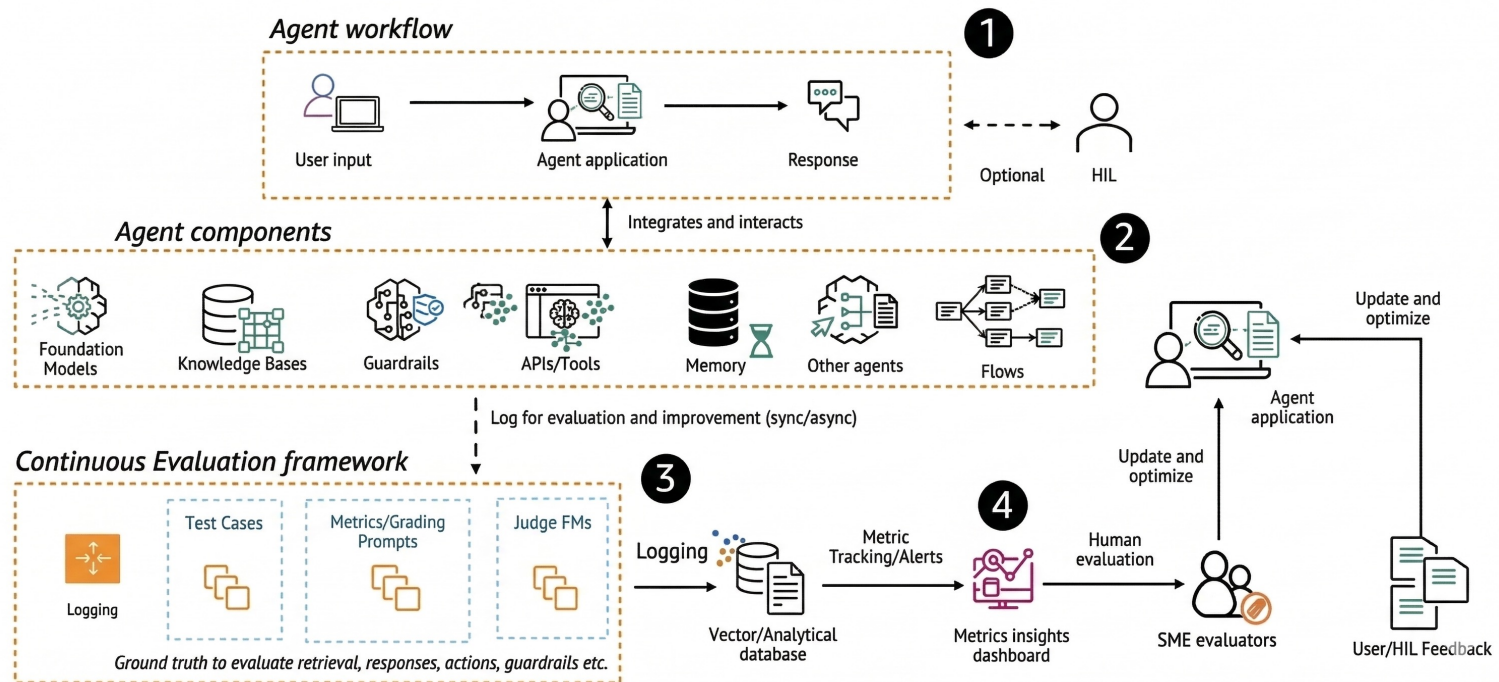
Agent Architectures



What tools do agents really need?



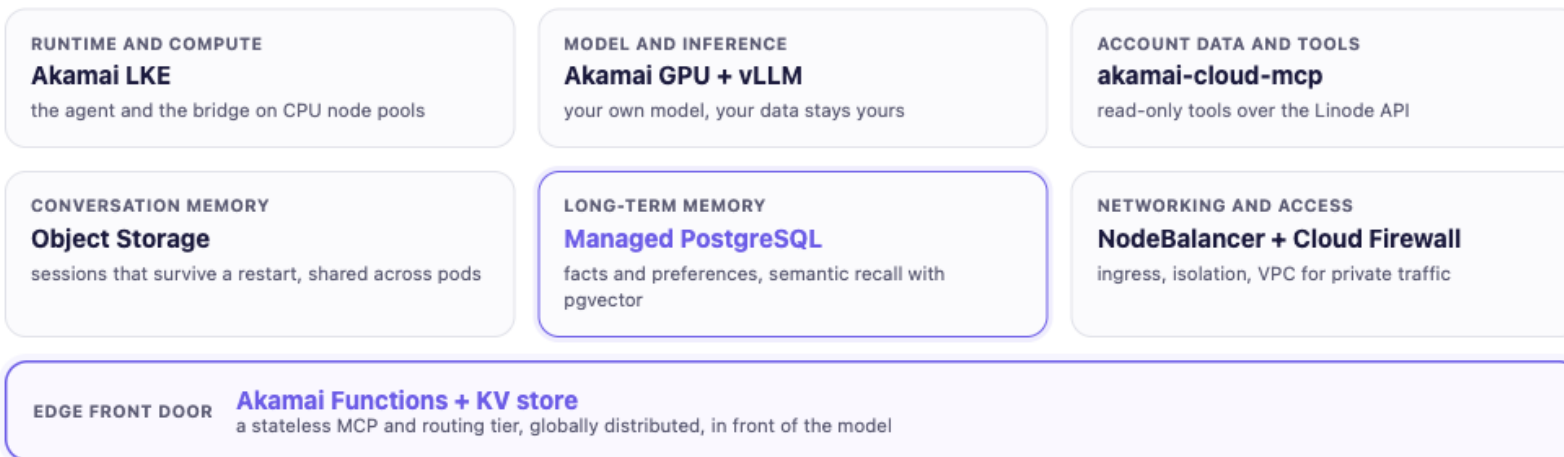
Components of an agentic system



Building agents on Akamai Cloud

Building AI Agents on The Akamai Cloud

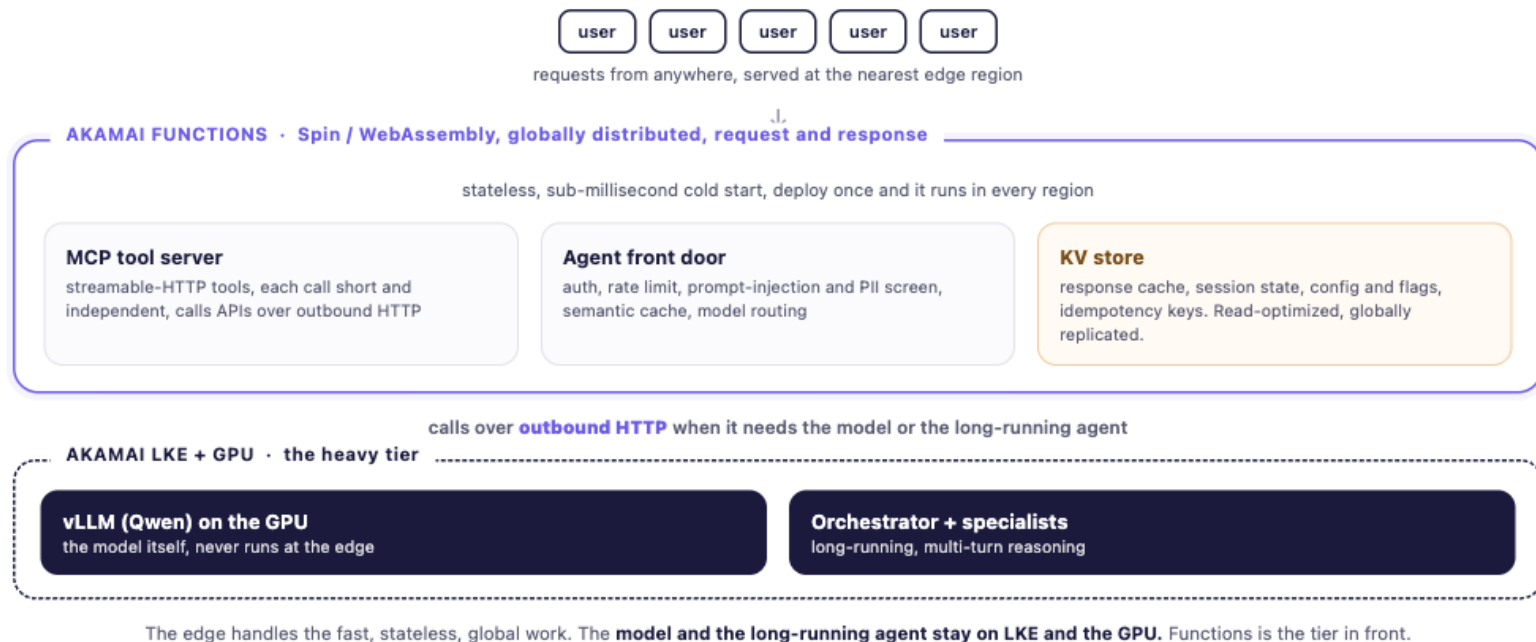
The agent, and every component around it, is an Akamai service.



The agent runs on Akamai. **So does everything around it: compute, GPU, data, memory, network, and the edge.**

Agents at the edge

AKAMAI FUNCTIONS · THE EDGE FRONT DOOR FOR AGENTS



Why Akamai: the whole stack is here

- Your own model on Akamai GPUs with vLLM. Your data stays yours.
- LKE for the agent and everything around it, on CPU nodes.
- Object Storage for conversation memory that survives a restart.
- Managed PostgreSQL with pgvector for long-term memory and recall.
- Akamai Functions to push pieces to the edge, close to users.

Run the open-source agent tools on Akamai too

Observe and remember

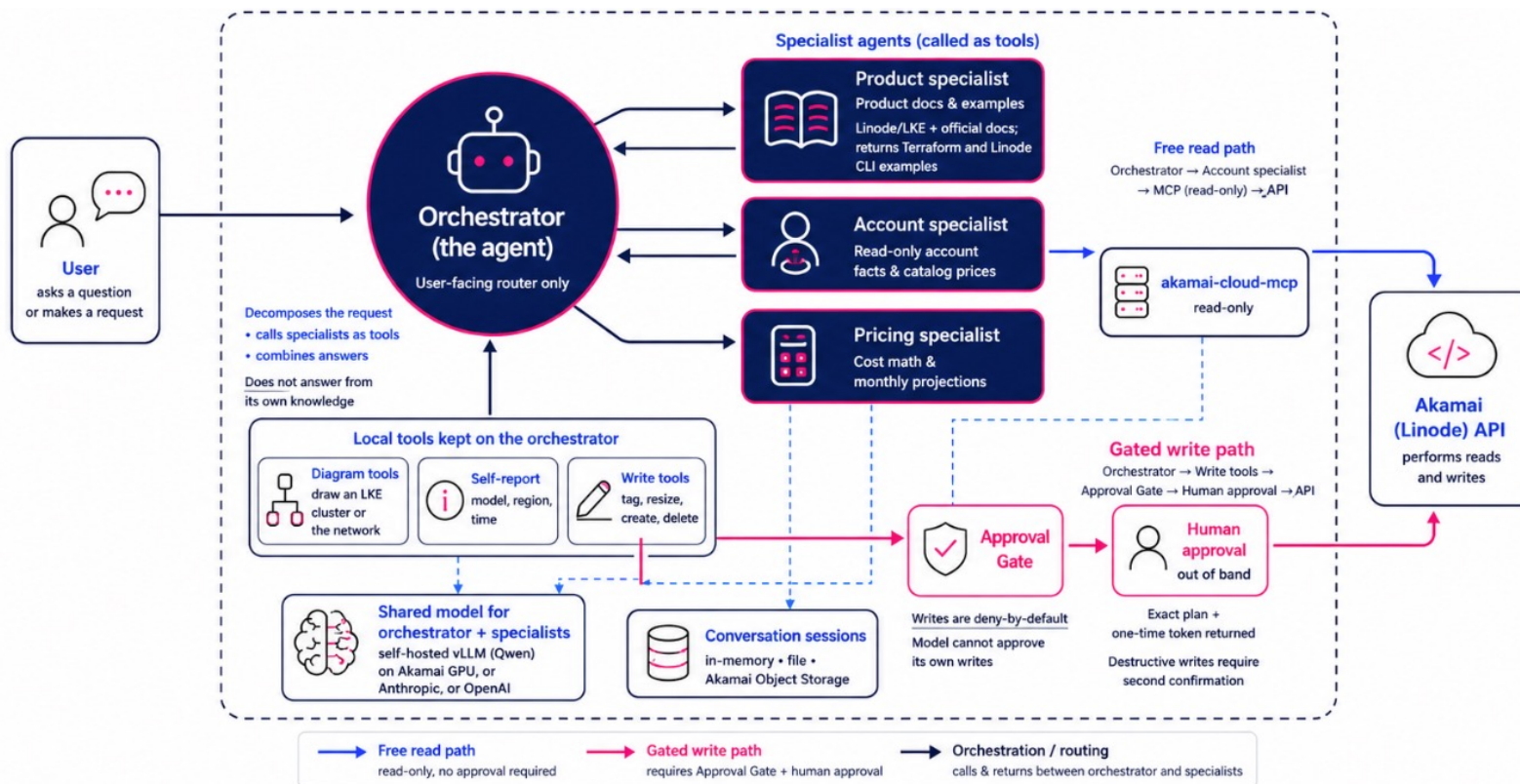
- **Observability**
 - Langfuse traces every run. On LKE, state in Managed PostgreSQL.
- **Memory**
 - mem0 or Letta. Vectors in Managed PostgreSQL with pgvector.
- **Vector and RAG**
 - pgvector, or Qdrant and Weaviate on LKE.

Inference and guardrails

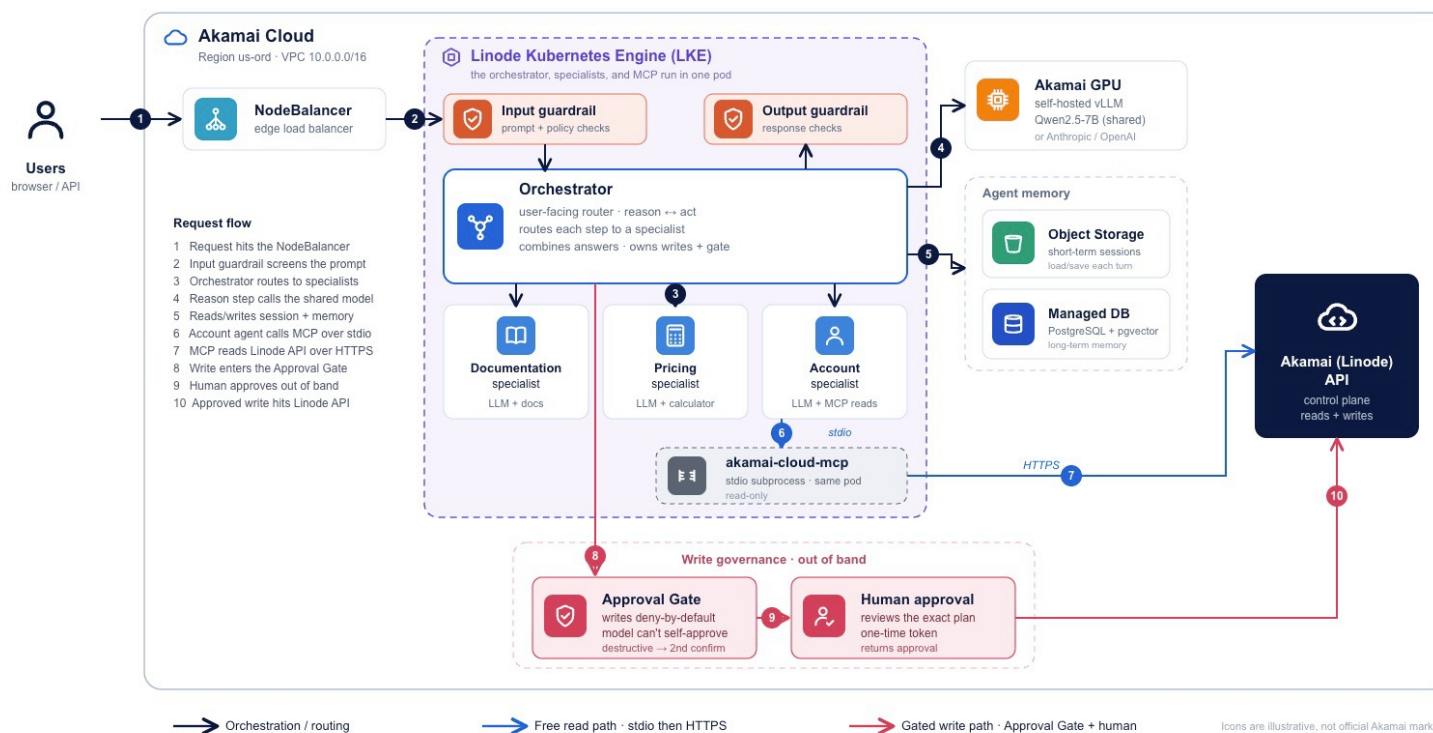
- **Serving inference and gateway.**
 - Ollama or vLLM (production) on GPU nodes
 - LiteLLM or agentgateway as the gateway.
- **Evals**
 - Promptfoo and Ragas in your LKE pipeline.
- **Guardrails**
 - NeMo Guardrails, or Llama Guard on vLLM.

The demo architecture

Demo: Akamai Solution Architect Agent

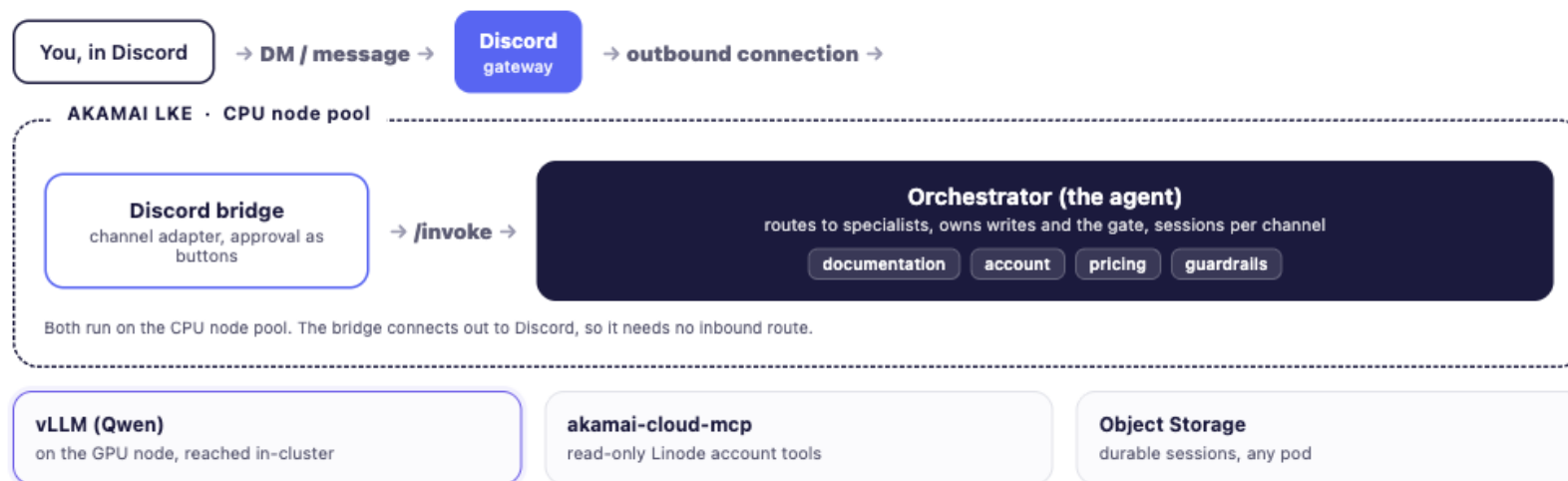


AI agents on Akamai Cloud



How it runs

THE AGENT, LIVE IN DISCORD · DEPLOYED ON AKAMAI LKE



A Discord message goes to the bridge, the bridge calls the agent, the agent reads your account and answers. **Writes pop up Approve and Decline buttons.**

The workshop: eight modules

01. First agent and tools

02. Real data with MCP

03. Guardrails

04. Memory and sessions

05. Diagrams that do not lie

06. Evals

07. Multi-agent

08. Deploy to LKE



It is all open. Go build.

Start with the platform, then the workshop.

<https://duan.bio/lunch-and-learn>

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